Project Scoping Document

Project Title: Supply Chain Risk & Margin Protection in Prestige Beauty via Enterprise Systems Integration

Project Timeline: June 2026 – September 2026

Date: June 12, 2026

1. Project Objectives & Boundaries

The primary objective of this project is to construct an analytical scenario-modeling framework evaluating the financial and operational impacts of transitioning a prestige beauty brand from a high-risk, single-source factory model to a diversified, dual-sourcing network. The project maps out how these insights operationalize safety stock configurations and protect luxury gross margins (70–80%) across core corporate systems: ERP (SAP S/4HANA), WMS (Manhattan Active), and Demand Planning software (Anaplan/o9 Solutions).

In-Scope: Feature engineering a dynamic Tariff Impact Coefficient by category; deconstructing supplier procurement vs. logistics delivery lead times; extracting Sephora consumer review velocity as an advanced demand signal; mapping output payloads directly to ERP Vendor Master database schemas.

Out-of-Scope: Deploying production-grade live API code injections into a client's active ecosystem; generalizable global macro-forecasting due to the 100-SKU baseline limit of the primary operational dataset.

2. Core Datasets & Sourcing Integration

To comprehensively solve the dual-sourcing problem, three distinct data layers are integrated:

Dataset 1: Primary Sourcing & Logistics Logs

Source: Kaggle Fashion & Beauty Supply Chain Dataset.
https://www.kaggle.com/datasets/harshsingh2209/supply-chain-analysis

Application: Establishes the baseline operational data for 100 beauty SKUs. It maps SKU-level manufacturing costs, historical supplier defect rates, carrier performance, and inventory volumes across Skincare, Cosmetics, and Haircare.

Variables Tracked: Product Type, SKU, Stock Levels, Order Quantities, Lead times, Lead time, Manufacturing Costs, Defect Rates, Inspection Results, Shipping Carriers, Transportation Modes, Routes, and Costs.

Dataset 2: Secondary Consumer Sell-Out Sentiment

Source: Kaggle Sephora Products and Customer Reviews Dataset.
https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews

Application: Serves as a forward-looking upstream demand signal. Review timestamp frequencies and rating momentum are used to forecast viral product spikes, ensuring demand software catches trends before warehouse stockouts occur.

Variables Tracked: product_name, brand_name, primary_category, secondary_category, tertiary_category, price_usd, ingredients, rating, reviews, loves_count, submission_time, is_recommended, skin_type, skin_tone, limited_edition, new, online_only, out_of_stock, sephora_exclusive.

Dataset 3: External Trade Policy & Freight Benchmarks (Additional Dataset)

Source: Office of the U.S. Trade Representative (USTR) Section 301 Schedules & European Commission Chapter 33 Customs Tariffs. [13, 14]

Application: Injected into the data curation layer to build a proxy Tariff Impact Coefficient. It bypasses the localized geography limitations of the primary dataset by mapping macro trade penalties directly to product composition:

- Cosmetics/Makeup: 25% East Asia packaging/component tariff penalty.
- Skincare: 10% European premium active ingredient/luxury glass premium.
- Haircare: 0% near-shore regional baseline.

3. Comprehensive Dashboard Design Plan

The datafolio dashboard will be built in Tableau or Power BI, structured across three highly interactive, role-specific tabs designed for executive S&OP alignment:

Tab 1: Executive S&OP Risk Matrix (Supply Chain Director View)

Visual Elements: A Sourcing Concentration Block Diagram tracking regional vendor nodes, color-coded heatmaps matching Defect Rates against Lead Time Variances, and a global vulnerability score by brand franchise.

Interactive Controls: A Global Tariff Sensitivity Slider (0% to 40%) allowing users to dynamically simulate how worsening trade friction impacts cross-border logistics lanes.

Tab 2: Margin Compression & Dual-Sourcing Simulation (CFO View)

Visual Elements: Waterfall charts showing the erosion of luxury gross margins under standard tariff penalties, paired with side-by-side scenario comparisons modeling a split volume allocation (e.g., 70% primary / 30% near-shore supplier).

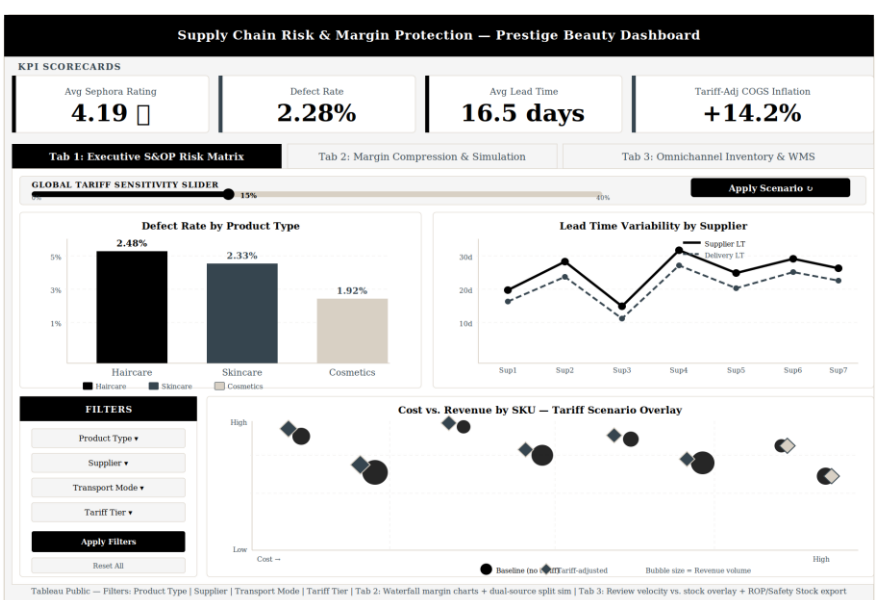
Core Metrics: Baseline Gross Margin % vs. Tariff-Adjusted Gross Margin %, total cost of goods sold (COGS) inflation, and margin recovery velocity.

Tab 3: Omnichannel Inventory & WMS Calibration (Warehouse Operations View)

Visual Elements: Dual-axis line charts overlaying Sephora customer review velocity against current warehouse stock levels to visually expose safety stock deficits.

System Integration Feature: An automated configuration window displaying dynamic Reorder Point (ROP) and Safety Stock parameter variables, formatted as an extractable data table optimized for direct entry into WMS master data.

Dashboard layout (wireframe):



4. Clear, Attainable Milestones & Timeline

The lifecycle breaks down into five sequential delivery phases:

[June 2, 2026 - Kickoff] —> [June 18 - Scoping] —> [July 2 - Curation] —> [July/Aug - EDA] —> [Aug - Final Delivery]

Milestone 1: Project Description and Scoping (Due June 18, 2026)

Deliverable: Formalization of the business problem, integration framework, technical boundaries, and rubric alignment.

Milestone 2: Data Curation & Feature Engineering (Due July 16, 2026)

Deliverable: Python pipeline development to merge Kaggle datasets; cleaning null review fields; executing the SQL category-based tariff risk proxy mapping logic.

Milestone 3: Exploratory Data Analysis Report

Due Date: Target: July 2026 (Date TBD)

Deliverable: Statistical distribution profiling of defect rates, variance tracking between the duplicate lead-time fields, and time-series analysis of Sephora review momentum.

Milestone 4: Interactive Dashboard Deployment

Due Date: Target: August 2026 (Date TBD)

Deliverable: Finalized multi-tab Tableau/Power BI interface containing functional tariff sliders and active inventory allocation parameter models.

Milestone 5: Final Portfolio Report

Due Date: Target: August 2026 (Date TBD)

Deliverable: A comprehensive written technical case study, public GitHub repository containing documented SQL/Python code blocks, and an executive slide deck.

5. Technical Stack & Generative AI Guardrails

Core Technologies: SQL (relational schema and metric calculation views); Python (pandas, numpy, scipy for optimization modeling); Tableau/Power BI (datafolio visualization).

GenAI Compliance: In compliance with the program's guidelines, Claude 4.6 Sonnet is utilized strictly as a development, code compilation, and layout partner. To maintain absolute transparency and integrity, all architectural parameters, domain logic, and system integration strategies are completely driven and verified by the author's 20+ years of professional ERP and technical experience.

6. Variable Relationships & Expected Analytical Findings

The following variable comparisons are expected to yield the most analytically significant relationships across the three datasets:

- Lead time variance (supplier_lead_time vs. delivery_lead_time) vs. Defect Rate: SKUs with the highest lead time instability are hypothesized to correlate with elevated defect rates, indicating supplier capacity stress.
- Manufacturing Cost (Tariff-Adjusted) vs. Gross Margin %: Applying the Tariff Impact Coefficient by product type is expected to show Skincare and Cosmetics categories experiencing the greatest margin compression relative to Haircare.

- Stock Levels vs. Sephora Review Velocity (submission_time trend): High-velocity review periods are expected to precede stockout events (out_of_stock = 1), validating the use of consumer sentiment as a leading inventory signal.
- Transportation Mode vs. Shipping Cost vs. Lead Time: Air transport is expected to show the lowest lead time but highest cost; Sea the inverse — surfacing the cost-speed tradeoff central to near-shoring decisions.
- Inspection Results (Pass/Fail/Pending) vs. Supplier Name: Defect concentration is expected to be non-uniform across suppliers, identifying which single-source nodes carry the highest quality risk.
- Sephora Price Tier vs. Rating Momentum (limited_edition, sephora_exclusive flags): Premium and exclusive SKUs are expected to show higher review velocity, informing which product franchises require the most supply chain resilience investment.

References

1. [1] Statista. (2024). Beauty & Personal Care — Worldwide. Gross margin benchmarks. https://www.statista.com/outlook/hc/beauty-personal-care/worldwide
2. [2] L'Oréal Finance. (2025). 2024 Annual Results. https://www.loreal-finance.com/eng/press-release/2024-annual-results
3. [3] LVMH. (2025). LVMH Annual Performance: Perfumes & Cosmetics; LIFE 360 traceability. https://www.lvmh.com/en/publications/lvmh-achieves-a-solid-performance-despite-an-unfavorable-global-economic-environment
4. [4] LVMH. (2025). A Purchasing Policy That Lives Up to Our Commitments. https://www.lvmh.com/en/commitment-in-action/a-purchasing-policy-that-lives-up-to-our-commitments
5. [5] The Estée Lauder Companies. (2025). FY2025 Q2 Earnings Release (Form 8-K). https://www.sec.gov/Archives/edgar/data/0001001250/000100125025000009/fy2025q2exhibit991.htm
6. [6] SAP SE. (2021). SAP Q4 2020 Quarterly Statement: L'Oréal as new S/4HANA customer. https://www.sec.gov/Archives/edgar/data/0001000184/000110465921009705/tm214658d1_ex99-1.htm
7. [7] Anaplan. (2024). Connected Planning: L'Oréal & Louis Vuitton SCM. https://www.anaplan.com/customers/loreal/
8. [8] o9 Solutions. (2025). How Estée Lauder Transforms Supply Chain. https://o9solutions.com/videos/how-estee-lauder-transforms-supply-chain-with-o9-solutions/
9. [9] Manhattan Associates. (2020). L'Oréal Global Deployment of Manhattan Active® WMS. GlobeNewswire. https://www.globenewswire.com/en/news-release/2020/10/06/2104281/9255/en
10. [10] APICS / ASCM. (2023). SCOR Digital Standard Framework. https://www.ascm.org/supply-chain-management-tools/scor-model/
11. [11] Singh, H. (2023). Supply Chain Analysis Dataset. Kaggle. https://www.kaggle.com/datasets/harshsingh2209/supply-chain-analysis
12. [12] Inky, N. (2023). Sephora Products and Skincare Reviews. Kaggle. https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews
13. [13] USTR. (2024). Section 301 Tariff Actions — Beauty Sector HS Codes 3303–3307. https://ustr.gov/issue-areas/enforcement/section-301-investigations/section-301-china/tariff-actions
14. [14] European Commission. MFN Tariff Rates — HS Chapter 33 Perfumery & Cosmetics. https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff_en